

n-Channel Polymer Field-Effect Transistors without Encapsulation Have over 21-Year Stability in Ambient Air

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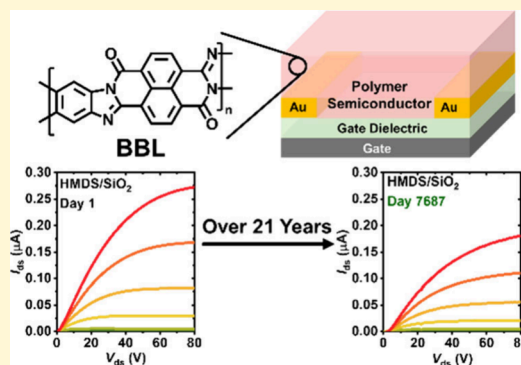


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Supporting Information

ABSTRACT: We report herein the ultrahigh stability of n-channel organic field-effect transistors (OFETs) based on ladder poly(benzimidazobenzophenanthroline) (BBL), without any encapsulation, which have remained fully functional after over 21 years of storage in ambient air, marking an unprecedented milestone for organic electronic devices. Although the BBL OFETs, fabricated in 2003 and 2011, had excellent functionality, the figures of merit slightly degraded from their original values. Thermal annealing of the aged OFETs enabled a complete reversal of the degradation process and full recovery of the electron transport properties. Trap density of states spectral analysis and characterization of OFET performance as a function of thermal annealing show that the ambient air-induced trapping sites accumulated at the semiconductor/dielectric interface over the long term are fully passivated. Our results provide evidence of over 21-year ambient air stability of BBL OFETs and offer insights into the potential design of next-generation semiconducting polymers and organic electronic devices with excellent stability.



Broad interest in developing organic electronic and optoelectronic devices for diverse applications has accelerated over the past few decades. Most notably are organic light-emitting diodes (OLEDs) for displays and lighting,^{1–4} organic field-effect transistors (OFETs) for flexible or wearable electronics,^{5–10} organic photovoltaic (OPV) cells for renewable power generation,^{11–19} and organic electrochemical transistors (OECTs) for biosensors^{20–22} and neuromorphic computing.^{23–25} Although considerable advances have been made in the performance metrics of these devices,^{6,26–43} their long-term air stability remains a major issue, which has not been addressed. We point out that the mobilities and threshold voltage of the earlier n-type polymer-based OFETs were rather low and high,^{44–48} respectively, compared to those of recent materials.^{48–54} Some recent studies have elucidated the mechanism of device degradation arising from bias stress instability, thermal stress, photo-degradation, and environmental factors such as water, oxygen, and their photo-oxidized products.^{55–64} In the case of n-type semiconducting polymers, which are highly susceptible to oxygen and water attacks partly due to their shallow lowest unoccupied molecular orbital (LUMO) energy level,^{64,65} little is understood regarding the extent of air stability of the materials and associated devices. Very few studies simultaneously reported high electron mobility ($>1 \text{ cm}^2/(\text{V s})$) and

ambient air stability for n-type semiconducting polymers.^{49–54} The majority of these reports either employed top-gate bottom-contact device geometry which provided a thick encapsulation layer against oxygen and water^{49,50,52,53} or examined the ambient air stability for a very short time (~ 30 – 100 days).^{49–54} As a result, many questions remain to be explored: What is the ambient air shelf life of an n-channel OFET? Which factors influence the rate of performance degradation upon exposure to air? Is it possible to reverse the ambient air-induced degradation process to fully recover the device performance?

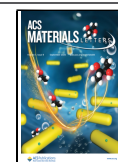
Herein we report an investigation of long-term ambient air stability of bottom-gate OFETs based on poly(benzimidazobenzophenanthroline) (BBL) (Figure 1a), which combines a deep LUMO energy level (-4.2 to -4.3 eV) with excellent electrochemical stability (Figure 1b). BBL enabled the first reported n-channel polymer OFETs in 2003.⁶ Fortunately, those early day BBL-based OFETs, fabricated in 2003 without

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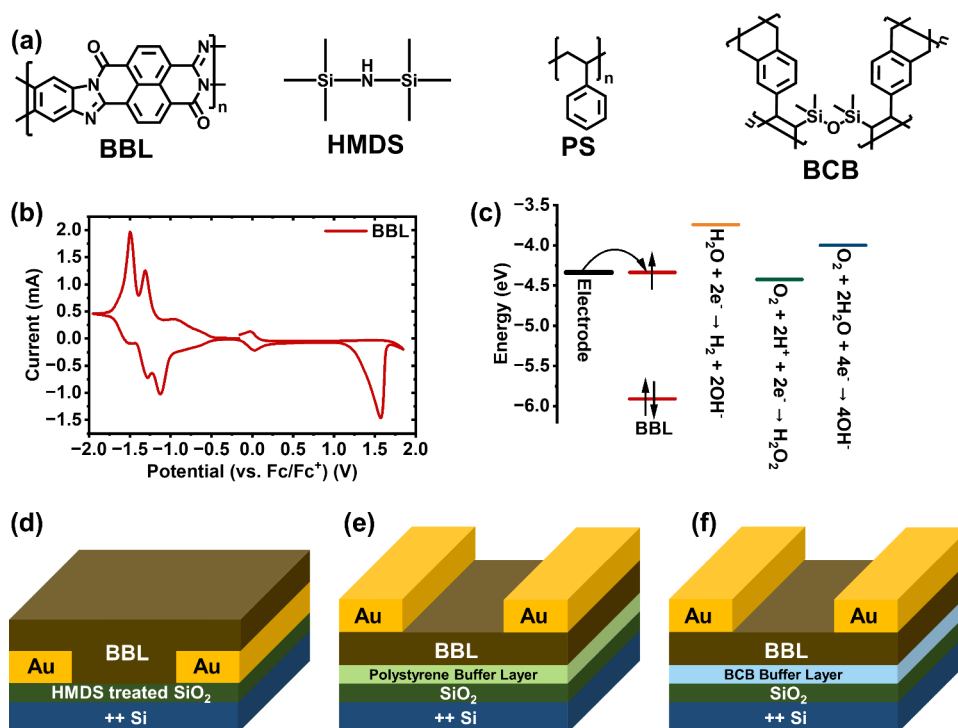


Figure 1. (a) Molecular structures of semiconducting polymer, poly(benzimidazobenzophenanthroline) (BBL), and different dielectric modifier materials: 1,1,1,3,3,3-hexamethyldisilazane (HMDS) self-assembly monolayer, polystyrene (PS) buffer layer, and divinylsilane-bis-benzocyclobutene (BCB) buffer layer. (b) Cyclic voltammogram of BBL thin film measured in 0.1 M tetrabutylammonium hexafluorophosphate (Bu_4NPF_6) in acetonitrile. (c) Thermodynamic consideration of potential degradation pathways of BBL upon exposure to oxygen and water in ambient air atmosphere. (d) Device architecture of bottom-gate bottom-contact BBL thin film transistors. (e, f) Device architecture of bottom-gate top-contact BBL thin film transistors with polystyrene dielectric (PS) buffer layer (e) and divinylsilane-bis-benzocyclobutene (BCB) dielectric buffer layer (f).

encapsulation, along with BBL-based OFETs fabricated in 2011 with different dielectric materials but also without encapsulation,⁶⁶ have provided a unique opportunity to investigate long-term ambient air stability and any degradation mechanism.

We found that the n-channel BBL OFETs remain operational with excellent saturation of drain current and good switching behavior even after more than 21 years of storage in ambient air, which is unprecedented for any organic electronic devices. Furthermore, we show that while the device geometry (i.e., bottom-gate bottom-contact (Figure 1d) versus bottom-gate top-contact (Figure 1e,f)) has marginal impacts on the long-term device stability, the retention rate of the charge transport properties depends on the choice of the dielectric materials. We show that the mechanism underlying the dielectric-dependent air stability of BBL OFETs is the variable concentration of water and oxygen accumulated at the dielectric/semiconductor interface. The more hydrophilic nature of the air-exposed divinylsilane-bis-benzocyclobutene (BCB) (Figure 1a) dielectric buffer layer compared to that of polystyrene (Figure 1a) dielectric buffer layer or 1,1,1,3,3,3-hexamethyldisilazane (HMDS) (Figure 1a) treated SiO₂ surface enables facile buildup of environmental impurities, which act as trapping sites resulting in a faster decay rate of electron transport properties. Moreover, we show that thermal annealing of the 21-year and 13-year aged BBL devices in an inert environment for a period of time can reverse any device degradation and completely recover the electron transport parameters to their original states seen in the freshly made (day-1) devices. This recovery can be rationalized by the

removal of adsorbed water and oxygen and passivation of those charge carrier trapping sites. Our study has thus demonstrated an unprecedented more than 21-year ultrahigh stability of BBL OFETs while providing valuable insights for the potential design of next-generation organic semiconductors and devices with excellent air stability.

■ AMBIENT AIR STABILITY OF n-Channel BBL OFETs

We characterized the bottom-gate BBL OFETs that were fabricated and studied in 2003 and 2011 to examine their long-term ambient air stability.^{6,66} Details on the fabrication and characterization procedures are summarized in the [Supporting Information](#) and described in the original reports.^{6,66} In short, the bottom-gate bottom-contact (BGBC) BBL transistors were fabricated on HMDS treated 300 nm thick SiO₂ surfaces (Figure 1d). The bottom-gate top-contact (BGTC) BBL transistors employed either BCB or PS as the polymer buffer layer on top of a 200 nm thick SiO₂ gate dielectric (Figure 1e,f). We investigated the effects of both the device geometry and the gate dielectric materials on the long-term ambient air stability of these BBL devices.

As shown in Figure S1a,c, the BGBC BBL OFETs showed excellent saturation of the drain current with more than 50% of the original values retained even after more than 21 years (7687 days) of sitting in ambient air. Furthermore, the transfer characteristics of these BBL transistors also exhibited linear behavior (Figure 2a, Figure S1b,d). As a result, the average saturated electron mobility (μ_e) of BBL after aging for more than 21 years in air was found to marginally decrease from $(8.2 \pm 1.1) \times 10^{-4}$ to $(3.0 \pm 2.4) \times 10^{-4} \text{ cm}^2/(\text{V s})$ (Table S1),

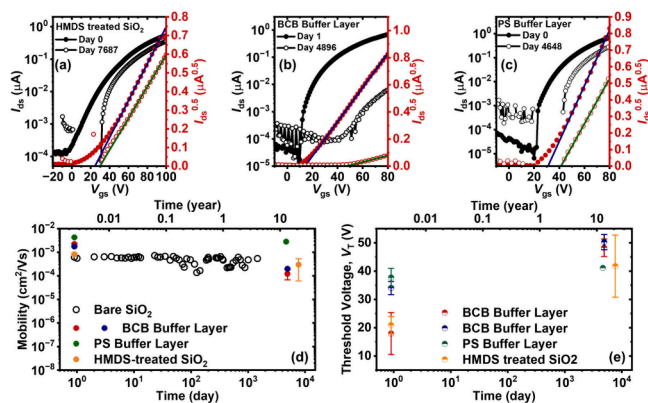


Figure 2. (a–c) Transfer curves of bottom-gate n-channel BBL thin film transistor devices of different dielectric layers after storage in ambient air atmosphere: (a) 1,1,1,3,3,3-hexamethylsilazane (HMDS) treated SiO₂; (b) divinylsilane-bis-benzocyclobutene (BCB) polymer buffer layer; and (c) polystyrene (PS) buffer layer. (d) Ambient air stability of charge carrier mobilities of BBL on different dielectric layers. Data for charge carrier mobility on bare SiO₂ are reproduced in part from ref 67. (e) Ambient air stability of threshold voltage (V_T) of bottom-gate BBL n-channel transistors fabricated on different dielectric layers.

indicating its remarkable air stability. We note that while these BGBC BBL OFETs were fabricated in 2003, they may not be the exact same devices that were used in the original report⁶ because many devices were fabricated in the course of optimization; thus, the absolute values of the electron mobility and threshold voltage may be different from those of our original study.⁶ Nevertheless, this result is an excellent extension of our previous work examining the air stability of n-channel BBL transistors on SiO₂ up to 4 years,⁶⁷ which is also reproduced in Figure 2d. However, the average threshold voltage (V_T) is found to increase by about 2-fold from 21 ± 3 to 42 ± 11 V (Table S1), which suggests increased concentrations of trap sites, whereas the on/off current ratios ($I_{\text{on}}/I_{\text{off}}$) decreased by an order of magnitude from 10^3 to 10^2 due to the higher off current in the 7687-day-old devices (Figure 2a). Nevertheless, the fact that these BBL OFETs are functional after more than two decades of sitting in ambient air is unprecedented and indicates the excellent air stability of BBL.

Excellent air stability was also observed for the BGTC BBL OFETs. Both devices using BCB and PS as the dielectric buffer layer can still operate with good current modulation (Figures 2b,c and S2), albeit with poorer electron transport parameters (Table S1), which is reasonable after more than 13 years of storage in ambient air atmosphere. In particular, the BCB devices exhibited about an order of magnitude decrease in μ_e from $(1.7\text{--}2.2) \times 10^{-3}$ to around 0.2×10^{-3} cm²/(V s) (Table S1) regardless of the original processing condition of the BCB layer. Moreover, the V_T value of the 4900-day-old BCB devices increased by at least 2- to 3-fold to about 50 V compared to the day-1 devices. The $I_{\text{on}}/I_{\text{off}}$ ratios decreased by about 2 orders of magnitude (Table S1). In the case of BBL OFETs using PS as the buffer layer, both μ_e and V_T were found to remain nearly constant at around $(3\text{--}4) \times 10^{-3}$ cm²/(V s) and 41 V, respectively, after 4648 days of storage in ambient air atmosphere. In contrast, the $I_{\text{on}}/I_{\text{off}}$ ratio of the PS devices decreased by nearly 2 orders of magnitude (Table S1) primarily due to the increased off current values (Figure 2c).

We note that the slightly lower μ_e observed in the HMDS-treated SiO₂ devices compared to the BCB and PS devices can be rationalized by multiple factors, including subtle differences in the film thickness of the BBL semiconducting layer,^{68,69} the dielectric constant of the buffer layer,⁶⁶ and contact resistance arisen from a coplanar versus staggered device architecture.^{68,69}

In addition, we found that the device geometry (i.e., bottom-contact versus top-contact) has minimal impacts on the storage stability of BBL transistors, as all devices remain functional with good current modulation and switching behavior, whereas the gate dielectric materials were found to strongly influence the air stability. Devices using PS as the buffer layer showed the highest retention rate of both μ_e and V_T , followed by HMDS treated SiO₂ devices and lastly the BCB devices (Figure 2d,e). The observed dependence of the air stability of BBL OFETs on dielectric materials is interesting, and the mechanism underlying this phenomenon is discussed below.

RECOVERY OF DEVICE PERFORMANCE OF BBL OFETs VIA TRAP PASSIVATION

We hypothesized that the observed dependence of the air stability of charge transport parameters of BBL transistor devices on the gate dielectric materials originates from the hydrophilicity/hydrophobicity of the dielectric/semiconductor interface. In particular, the more hydrophilic nature of the air-exposed BCB could enable water and oxygen to accumulate at the semiconductor/dielectric interface upon diffusion into the BBL films, whereas the more hydrophobic polystyrene films and HMDS self-assembled monolayers likely hinder accumulation of water and oxygen. We note that there is also the possibility of oxidation or hydrolysis of the BCB layer in air. Such environmental impurities (i.e., O₂ and H₂O) would result in a higher trap concentration and the faster decay rates of μ_e , V_T , and $I_{\text{on}}/I_{\text{off}}$ seen in BCB devices compared to those of PS devices and HMDS treated devices.

To test this hypothesis, we first evaluated the thermodynamics and kinetics of oxygen and water diffusion into the semiconducting BBL films by comparing their kinetic diameters to the solid-state packing distances of the polymer crystallites. Our previous study examining the chain length dependent thin film microstructures of BBL showed that the lamellar packing distances (d_{lam}) are around 8 Å, whereas the π - π spacings ($d_{\pi-\pi}$) are around 3.3–3.4 Å, both of which are independent of the polymer molecular weight (Figure S3).⁷⁰ Thus, while there is a kinetic barrier for oxygen diffusion along the π -stacking direction ($d_{\text{oxygen}} = 3.46 \text{ Å} > d_{\pi-\pi}$), O₂ molecules can freely enter the polymer crystallites along the lamellar axis. Furthermore, the relatively small diameter of water ($d_{\text{water}} = 2.65 \text{ Å}$) suggests that it can easily penetrate the polymer crystallites from both the lamellar and π -stacking directions. In addition, the amorphous regions of the BBL films are also susceptible to oxygen and water accumulation over time. As shown in Figure 1c, the coexistence of these environmental factors can lead to undesirable side reactions resulting in trap sites, thereby lowering μ_e while increasing V_T .

In an attempt to recover the performance of BBL transistors to their original state, we thermally annealed those 13-year-old devices at 170 °C in an inert environment to remove O₂ and H₂O and passivate the trap sites. The output and transfer curves of BBL transistors on the BCB buffer layer with varying annealing time are shown in Figures 3 and 4a,b, respectively, while those of the PS devices are demonstrated in Figure S4 and Figure 4c,d. The annealing time dependent electron

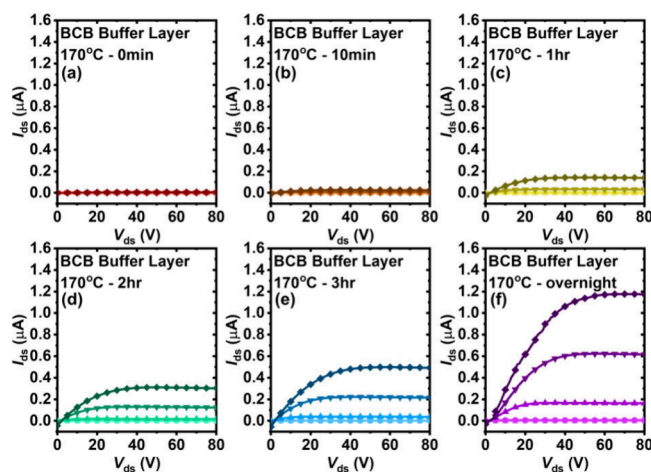


Figure 3. Effects of annealing time on the output curves of n-channel BBL organic field-effect transistors with BCB as the dielectric buffer layer: (a) 0, (b) 10, (c) 60, (d) 120, (e) 180, and (f) 1320 min. Channel dimensions were $100\ \mu\text{m}$ (L) $\times$ $1000\ \mu\text{m}$ (W).

transport properties of BBL transistors are illustrated in Figure 4e–g for the BCB buffer layer and Figure 4h–j for the PS buffer layer.

As shown in Figure 3 and Figure S4, output curves of both the BCB and PS devices exhibited enhanced saturation of the drain current with longer annealing time. The transfer curves of these BBL transistor devices were also found to gradually left shift to lower gate voltage (V_{gs}) values as the annealing time increased (Figure 4a–d), suggesting a lower operating voltage. Furthermore, the relationship between the square root of the drain current ($I_{ds}^{0.5}$) and V_{gs} , especially in the BCB devices, transitioned from an upward curve to a highly linear line (Figure 4b), indicating decreased energetic disorder.⁷¹ As a result, the average μ_e value of the 13-year-old BCB devices was fully recovered from $(0.12 \pm 0.06) \times 10^{-3}\ \text{cm}^2/(\text{V s})$ to its as-fabricated value of $(2.1 \pm 0.2) \times 10^{-3}\ \text{cm}^2/(\text{V s})$ after 120 min of annealing at $170\ ^\circ\text{C}$ in a nitrogen-filled glovebox (Tables S2 and S3, Figure 4e). The μ_e was further improved to $(5.1 \pm 0.3) \times 10^{-3}\ \text{cm}^2/(\text{V s})$ after overnight annealing (1320 min) at the same condition (Table S2). A similar trend was also observed for the PS devices whereby μ_e was completely recovered to the day-1 value of $5.9 \times 10^{-3}\ \text{cm}^2/(\text{V s})$ after 10 min of annealing at $170\ ^\circ\text{C}$ in a N_2 -filled environment (Table S4 and Figure 4h). A longer annealing time resulted in a much improved charge carrier mobility approaching $1.4 \times 10^{-2}\ \text{cm}^2/(\text{V s})$ (Table S4). The observed complete recovery of electron mobility indicates a lower degree of charge carrier trapping, which can be rationalized by the expulsion of adsorbed water and oxygen at the dielectric/semiconductor interface.

The V_T of both the BCB and PS devices was found to rapidly decrease toward and even beyond the original (day-1) benchmark once the devices were annealed for more than 180 min despite the seemingly initial resistance to the thermal annealing method (Figure 4f,i). The observed progressive decrease of V_T upon heat treatment can be explained by the gradual removal of those air-induced trap sites, thus validating our proposed hypothesis. In addition, the I_{on}/I_{off} ratios of both the BCB and PS devices increased by 1–2 orders of magnitude, approaching their original state ($I_{on}/I_{off} \sim 10^4$) seen in freshly made devices (Figure 4g,j), which also corroborates our hypothesis.

Moreover, we found that the PS devices recovered at a faster rate, whereby the crossover point of the original values occurred at a relatively shorter annealing time compared to that of the BCB devices (Figure 4e–j). This finding confirms that the hydrophilic nature of the air-exposed BCB layer enables a higher concentration of water and oxygen accumulation at the semiconductor/dielectric interface, therefore requiring a longer annealing time to completely remove them. Our results have thus showed that oxygen, water, and their associated products act as charge trapping sites and are responsible for the lower electron transport properties of BBL transistors upon long-term exposure to ambient air and that thermal annealing in an inert environment for a period of time is an effective strategy to desorb those impurities and decrease the overall concentrations of traps.

TRAP DENSITY OF STATES ANALYSIS

We employed Gr̈unewald's method to visualize the trap density of states (DOS)^{72,73} of the BBL transistor devices as a function of annealing time to gain more insights into how thermal annealing influences the trap passivation process. We found that the interfacial potential (V_0) gradually increased to higher values with increasing heat treatment time (Figure S5a,b). This suggests that the energetic distribution of trap sites shifted further away from the Fermi energy and potentially the transport edge, thus agreeing with the observed recovery and enhancement of electron transport properties. The trap DOS spectra of both BCB and PS devices were found to strongly depend on the annealing time (Figure S5c,d). In particular, the distribution of traps downshifted as the annealing time increased, indicating an overall lower trap concentration.⁷³ These findings are in excellent agreement with the observed improved μ_e , higher I_{on}/I_{off} ratio, and lower V_T .

IMPLICATIONS FOR THE DESIGN OF NEXT-GENERATION AMBIENT AIR STABLE OFET AND OECT MATERIALS AND DEVICES

We now provide some design strategies for next-generation n-type semiconducting materials and devices that can achieve the “improved properties, enhanced device performance, and rugged air stability” trifecta. Our results have demonstrated that the combination of the high electron affinity of BBL ($EA \sim 4.2$ – $4.3\ \text{eV}$) and the tight π - π stacking distances enables BBL n-channel OFETs to remain operational even after more than 21 years of exposure to ambient air. Such a device longevity has yet to be demonstrated for any other π -conjugated polymers.

From a molecular engineering perspective, a key requirement to ensure tightly packed polymer crystallites to prevent oxygen and water penetration is a rigid-rod polymer chain topology without a long alkyl side. Incorporation of strongly electron-deficient substituents including fluorine,^{74–77} chlorine,^{76–79} or cyano^{80–83} directly onto the ladder polymer backbone is an effective design strategy to deepen the LUMO energy level, which has been successfully demonstrated for several classes of π -conjugated polymers.^{74–83} Transition from a donor–acceptor copolymer motif to an acceptor–acceptor homopolymer design can also lead to a lower lying LUMO energy level.^{84,85} Furthermore, engineering several redox-active sites (e.g., hydroxyl, carbonyl, imine, etc.) along the polymer's π -conjugation pathway is another method to downshift the LUMO energy level.⁸⁶ In addition, protonation⁸⁷ or

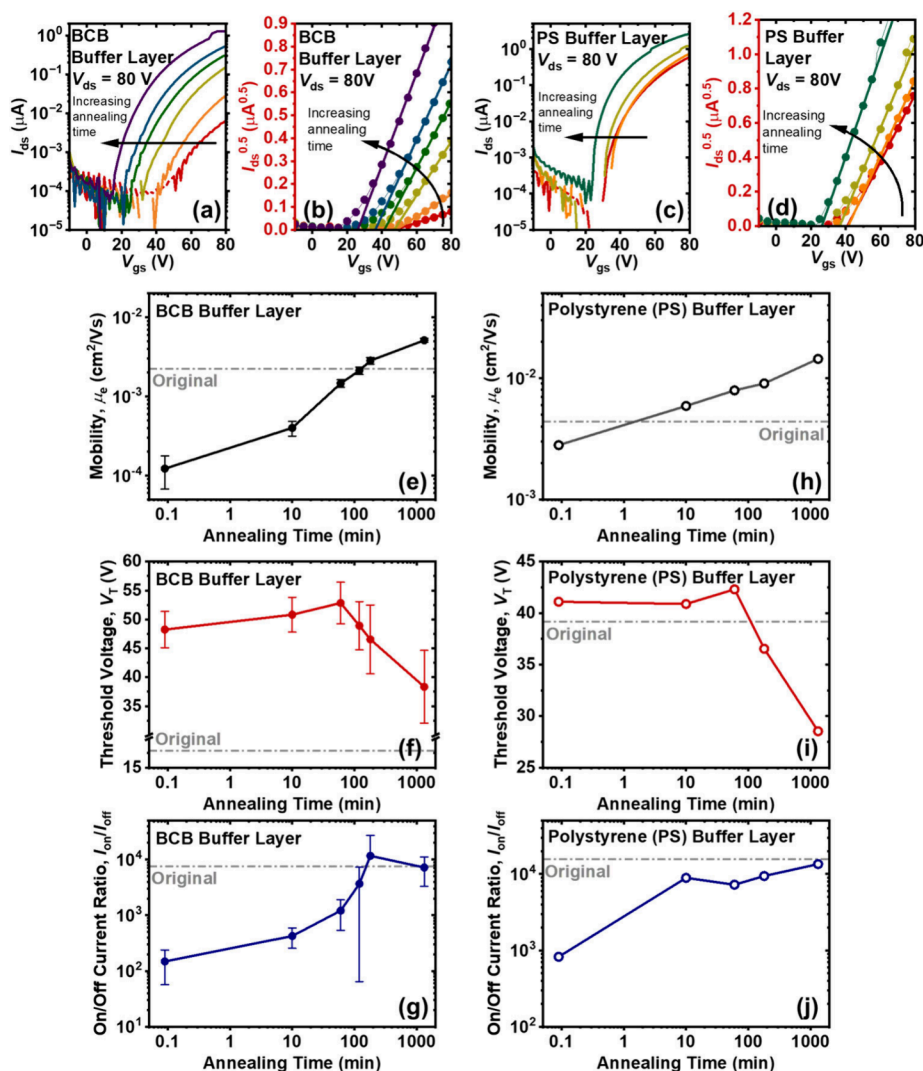


Figure 4. (a–d) Effects of annealing time on the transfer curves of bottom-gate top-contact n-channel BBL organic field-effect transistors with BCB as the dielectric buffer layer (a, b) and PS as the dielectric layer (c, d). (e–g) Effects of annealing time on electrical parameters of BBL transistors with BCB buffer layer: (e) electron mobility, (f) threshold voltage, (g) on/off current ratio. (h–j) Effects of annealing time on electrical parameters of BBL transistors with polystyrene (PS) buffer layer: (h) electron mobility, (i) threshold voltage, and (j) on/off current ratio.

quaternization⁸⁸ of such redox-active sites has also been shown to pull the electron density away from the backbone and increase the EA.^{87,88} Moreover, theoretical calculations via density functional theory (DFT) calculations and even molecular dynamics (MD) simulations should be performed to ensure that the ladder polymer backbone remains highly planar with close packing distances upon incorporation of these design strategies.

From a device engineering point of view, suppressing the accumulation of water and oxygen at the dielectric/semiconductor interface is imperative and can be achieved by using more hydrophobic dielectric buffer layers. Additional encapsulation layers deposited on top of the semiconducting polymer layers can also be considered for bottom-gate devices. Furthermore, a thick film of Cytop or poly(methyl methacrylate) (PMMA) above the semiconducting layer could simultaneously act as a gate dielectric material and provide a barrier against the penetration of undesirable impurities.

By recharacterizing and investigating n-channel BBL OFETs of different device architectures fabricated in 2003 and 2011, without encapsulation, and stored in ambient air, we have gained unique insights into factors that affect the long-term ambient air stability and the degradation mechanism of organic electronic devices. Although the 21-year and 13-year-old OFETs still had excellent functionality, the electron mobility and current on/off ratio decreased by an order of magnitude while the threshold voltage increased by about 2-fold compared to their original (day-1) performance. We found that thermal annealing of the BBL OFETs in an inert atmosphere completely recovered their original performance and thus verified our hypothesis that O_2 and H_2O from ambient air had diffused into the organic semiconductors and formed charge traps at the semiconductor/insulator interface over the long term. Furthermore, trap density of states spectral analysis showed that the original choice of the gate dielectric materials (hydrophobic or hydrophilic) influenced the rate of long-term degradation of the OFET devices and rate of recovery by thermal annealing.

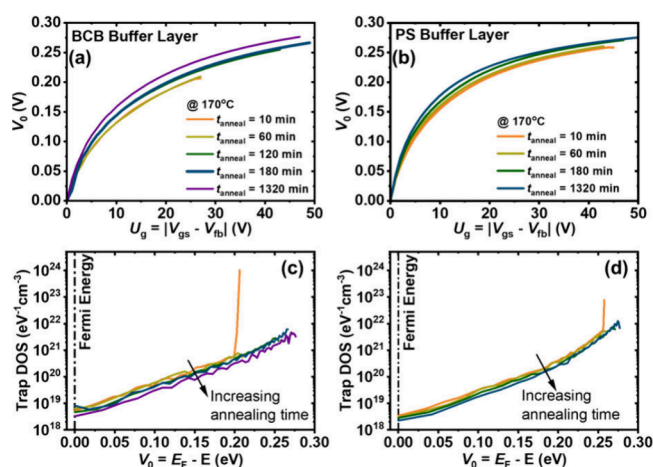


Figure 5. (a, b) Gate voltage dependent interface potential of bottom-gate top-contact BBL field-effect transistors using BCB buffer layer (a) and PS buffer layer (b). (c, d) Trap density of states (DOS) of BBL transistors as a function of annealing time.

Our results thus provide evidence of over 21-year stability of BBL-based n-channel OFETs in ambient air while also giving unique insights into the potential design of next-generation semiconducting polymers and organic electronic devices with excellent stability.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.5c00785>.

Details of materials, fabrication procedure of bottom-gate bottom-contact OFET devices, fabrication procedure of bottom-gate top-contact OFET devices, output and transfer curves of OFET devices, summary of ambient air stability of electrical parameters of OFET devices fabricated using different dielectric buffer layers, and procedure to calculate trap density of states (DOS) by Grünwald's method (PDF)

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Author Contributions

D.K.T. characterized the ambient air stability of BBL field-effect transistors, analyzed trap density of states (DOS), and performed data visualization. F.S.K. fabricated and charac-

terized the original (2011) bottom-gate top-contact BBL field-effect transistors. S.A.J. supervised all aspects of the studies. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. CRediT: **Duyen K. Tran** conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing - original draft, writing - review & editing; **Felix Sunjoo Kim** data curation, formal analysis, investigation; **Samson A. Jenekhe** conceptualization, funding acquisition, project administration, supervision, writing - original draft, writing - review & editing.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Tang, C. W.; VanSlyke, S. A. Organic electroluminescent diodes. *Appl. Phys. Lett.* **1987**, *51*, 913–915.
- (2) Burroughes, J. H.; Bradley, D. D. C.; Brown, A. R.; Marks, R. N.; Mackay, K.; Friend, R. H.; Burns, P. L.; Holmes, A. B. Light-emitting diodes based on conjugated polymers. *Nature* **1990**, *347*, 539–541.
- (3) Kulkarni, A. P.; Tonzola, C. J.; Babel, A.; Jenekhe, S. A. Electron Transport Materials for Organic Light-Emitting Diodes. *Chem. Mater.* **2004**, *16*, 4556–4573.
- (4) Gather, M. C.; Köhnen, A.; Meerholz, K. White Organic Light-Emitting Diodes. *Adv. Mater.* **2011**, *23*, 233–248.
- (5) Sirringhaus, H.; Brown, P. J.; Friend, R. H.; Nielsen, M. M.; Bechgaard, K.; Langeveld-Voss, B. M. W.; Spiering, A. J. H.; Janssen, R. A. J.; Meijer, E. W.; Herwig, P.; de Leeuw, D. M. Two-dimensional charge transport in self-organized, high-mobility conjugated polymers. *Nature* **1999**, *401*, 685–688.
- (6) Babel, A.; Jenekhe, S. A. High Electron Mobility in Ladder Polymer Field-Effect Transistors. *J. Am. Chem. Soc.* **2003**, *125*, 13656–13657.
- (7) Yan, H.; Chen, Z.; Zheng, Y.; Newman, C.; Quinn, J. R.; Dotz, F.; Kastler, M.; Facchetti, A. A high-mobility electron-transporting polymer for printed transistors. *Nature* **2009**, *457*, 679–686.
- (8) Sirringhaus, H. 25th Anniversary Article: Organic Field-Effect Transistors: The Path Beyond Amorphous Silicon. *Adv. Mater.* **2014**, *26*, 1319–1335.
- (9) Chortos, A.; Bao, Z. Skin-inspired electronic devices. *Mater. Today* **2014**, *17*, 321–331.
- (10) Kim, D.; Kim, F. S. Materials Chemistry, Device Engineering, and Promising Applications of Polymer Transistors. *Chem. Mater.* **2021**, *33*, 7572–7594.
- (11) Tang, C. W. Two-layer organic photovoltaic cell. *Appl. Phys. Lett.* **1986**, *48*, 183–185.
- (12) Yu, G.; Gao, J.; Hummelen, J. C.; Wudl, F.; Heeger, A. J. Polymer Photovoltaic Cells: Enhanced Efficiencies via a Network of Internal Donor-Acceptor Heterojunctions. *Science* **1995**, *270*, 1789–1791.
- (13) Halls, J. J. M.; Walsh, C. A.; Greenham, N. C.; Marseglia, E. A.; Friend, R. H.; Moratti, S. C.; Holmes, A. B. Efficient photodiodes from interpenetrating polymer networks. *Nature* **1995**, *376*, 498.
- (14) Heeger, A. J. 25th Anniversary Article: Bulk Heterojunction Solar Cells: Understanding the Mechanism of Operation. *Adv. Mater.* **2014**, *26*, 10.
- (15) Lee, C.; Lee, S.; Kim, G.-U.; Lee, W.; Kim, B. J. Recent Advances, Design Guidelines, and Prospects of All-Polymer Solar Cells. *Chem. Rev.* **2019**, *119*, 8028–8086.

- (16) Zhang, G.; Lin, F. R.; Qi, F.; Heumüller, T.; Distler, A.; Egelhaaf, H.-J.; Li, N.; Chow, P. C. Y.; Brabec, C. J.; Jen, A. K. Y.; Yip, H.-L. Renewed Prospects for Organic Photovoltaics. *Chem. Rev.* **2022**, *122*, 14180–14274.
- (17) Yan, C.; Barlow, S.; Wang, Z.; Yan, H.; Jen, A. K. Y.; Marder, S. R.; Zhan, X. Non-fullerene acceptors for organic solar cells. *Nature Reviews Materials* **2018**, *3*, 18003.
- (18) Yao, H.; Hou, J. Recent Advances in Single-Junction Organic Solar Cells. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202209021.
- (19) Yi, J.; Zhang, G.; Yu, H.; Yan, H. Advantages, challenges and molecular design of different material types used in organic solar cells. *Nat. Rev. Mater.* **2024**, *9*, 46–62.
- (20) Rivnay, J.; Owens, R. M.; Malliaras, G. G. The Rise of Organic Bioelectronics. *Chem. Mater.* **2014**, *26*, 679–685.
- (21) Rivnay, J.; Inal, S.; Salleo, A.; Owens, R. M.; Berggren, M.; Malliaras, G. G. Organic electrochemical transistors. *Nat. Rev. Mater.* **2018**, *3*, 17086.
- (22) Marks, A.; Griggs, S.; Gasparini, N.; Moser, M. Organic Electrochemical Transistors: An Emerging Technology for Biosensing. *Adv. Mater. Interfaces* **2022**, *9*, 2102039.
- (23) van de Burgt, Y.; Melianas, A.; Keene, S. T.; Malliaras, G.; Salleo, A. Organic electronics for neuromorphic computing. *Nat. Electron.* **2018**, *1*, 386–397.
- (24) Gkoupidenis, P.; Zhang, Y.; Kleemann, H.; Ling, H.; Santoro, F.; Fabiano, S.; Salleo, A.; van de Burgt, Y. Organic mixed conductors for bioinspired electronics. *Nat. Rev. Mater.* **2024**, *9*, 134–149.
- (25) van de Burgt, Y.; Lubberman, E.; Fuller, E. J.; Keene, S. T.; Faria, G. C.; Agarwal, S.; Marinella, M. J.; Alec Talin, A.; Salleo, A. A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nat. Mater.* **2017**, *16*, 414–418.
- (26) Earmme, T.; Ahmed, E.; Jenekhe, S. A. Solution-Processed Highly Efficient Blue Phosphorescent Polymer Light-Emitting Diodes Enabled by a New Electron Transport Material. *Adv. Mater.* **2010**, *22*, 4744–4748.
- (27) Earmme, T.; Jenekhe, S. A. High-performance multilayered phosphorescent OLEDs by solution-processed commercial electron-transport materials. *J. Mater. Chem.* **2012**, *22*, 4660–4668.
- (28) Kim, F. S.; Park, C. H.; Na, Y.; Jenekhe, S. A. Effects of ladder structure on the electronic properties and field-effect transistor performance of Poly(benzobisimidazobenzophenanthroline). *Org. Electron.* **2019**, *69*, 301–307.
- (29) Venkateshvaran, D.; Nikolka, M.; Sadhanala, A.; Lemaire, V.; Zelazny, M.; Kepa, M.; Hurhangee, M.; Kronemeijer, A. J.; Pecunia, V.; Nasrallah, I.; Romanov, I.; Broch, K.; McCulloch, I.; Emin, D.; Olivier, Y.; Cornil, J.; Beljonne, D.; Sirringhaus, H. Approaching disorder-free transport in high-mobility conjugated polymers. *Nature* **2014**, *515*, 384–388.
- (30) Fratini, S.; Nikolka, M.; Salleo, A.; Schweicher, G.; Sirringhaus, H. Charge transport in high-mobility conjugated polymers and molecular semiconductors. *Nat. Mater.* **2020**, *19*, 491–502.
- (31) Li, H.; Earmme, T.; Ren, G.; Saeki, A.; Yoshikawa, S.; Murari, N. M.; Subramanian, S.; Crane, M. J.; Seki, S.; Jenekhe, S. A. Beyond Fullerenes: Design of Nonfullerene Acceptors for Efficient Organic Photovoltaics. *J. Am. Chem. Soc.* **2014**, *136*, 14589–14597.
- (32) Hwang, Y.-J.; Courtright, B. A. E.; Ferreira, A. S.; Tolbert, S. H.; Jenekhe, S. A. 7.7% Efficient All-Polymer Solar Cells. *Adv. Mater.* **2015**, *27*, 4578–4584.
- (33) Kolhe, N. B.; Tran, D. K.; Lee, H.; Kuzuhara, D.; Yoshimoto, N.; Koganezawa, T.; Jenekhe, S. A. New Random Copolymer Acceptors Enable Additive-Free Processing of 10.1% Efficient All-Polymer Solar Cells with Near-Unity Internal Quantum Efficiency. *ACS Energy Lett.* **2019**, *4*, 1162–1170.
- (34) Zhao, W.; Qian, D.; Zhang, S.; Li, S.; Inganäs, O.; Gao, F.; Hou, J. Fullerene-Free Polymer Solar Cells with over 11% Efficiency and Excellent Thermal Stability. *Adv. Mater.* **2016**, *28*, 4734–4739.
- (35) Yuan, J.; Zhang, Y.; Zhou, L.; Zhang, G.; Yip, H.-L.; Lau, T.-K.; Lu, X.; Zhu, C.; Peng, H.; Johnson, P. A.; Leclerc, M.; Cao, Y.; Ulanski, J.; Li, Y.; Zou, Y. Single-Junction Organic Solar Cell with over 15% Efficiency Using Fused-Ring Acceptor with Electron-Deficient Core. *Joule* **2019**, *3*, 1140–1151.
- (36) Zhu, L.; Zhang, M.; Xu, J.; Li, C.; Yan, J.; Zhou, G.; Zhong, W.; Hao, T.; Song, J.; Xue, X.; Zhou, Z.; Zeng, R.; Zhu, H.; Chen, C.-C.; MacKenzie, R. C. I.; Zou, Y.; Nelson, J.; Zhang, Y.; Sun, Y.; Liu, F. Single-junction organic solar cells with over 19% efficiency enabled by a refined double-fibril network morphology. *Nat. Mater.* **2022**, *21*, 656–663.
- (37) Zheng, Z.; Wang, J.; Bi, P.; Ren, J.; Wang, Y.; Yang, Y.; Liu, X.; Zhang, S.; Hou, J. Tandem Organic Solar Cell with 20.2% Efficiency. *Joule* **2022**, *6*, 171–184.
- (38) Huang, W.; Chen, J.; Yao, Y.; Zheng, D.; Ji, X.; Feng, L.-W.; Moore, D.; Glavin, N. R.; Xie, M.; Chen, Y.; Pankow, R. M.; Surendran, A.; Wang, Z.; Xia, Y.; Bai, L.; Rivnay, J.; Ping, J.; Guo, X.; Cheng, Y.; Marks, T. J.; Facchetti, A. Vertical organic electrochemical transistors for complementary circuits. *Nature* **2023**, *613*, 496–502.
- (39) Kukhta, N. A.; Marks, A.; Luscombe, C. K. Molecular Design Strategies toward Improvement of Charge Injection and Ionic Conduction in Organic Mixed Ionic-Electronic Conductors for Organic Electrochemical Transistors. *Chem. Rev.* **2022**, *122*, 4325–4355.
- (40) Sun, H.; Vagin, M.; Wang, S.; Crispin, X.; Forchheimer, R.; Berggren, M.; Fabiano, S. Complementary Logic Circuits Based on High-Performance n-Type Organic Electrochemical Transistors. *Adv. Mater.* **2018**, *30*, 1704916.
- (41) Wu, H.-Y.; Yang, C.-Y.; Li, Q.; Kolhe, N. B.; Strakosas, X.; Stoessel, M.-A.; Wu, Z.; Jin, W.; Savvakis, M.; Kroon, R.; Tu, D.; Woo, H. Y.; Berggren, M.; Jenekhe, S. A.; Fabiano, S. Influence of Molecular Weight on the Organic Electrochemical Transistor Performance of Ladder-Type Conjugated Polymers. *Adv. Mater.* **2022**, *34*, 2106235.
- (42) Guo, J.; Flagg, L. Q.; Tran, D. K.; Chen, S. E.; Li, R.; Kolhe, N. B.; Giridharagopal, R.; Jenekhe, S. A.; Richter, L. J.; Ginger, D. S. Hydration of a Side-Chain-Free n-Type Semiconducting Ladder Polymer Driven by Electrochemical Doping. *J. Am. Chem. Soc.* **2023**, *145*, 1866–1876.
- (43) Thuc-Quyen, N. Insight into degradation mechanisms of bulk heterojunction organic solar cells. In *Organic, Hybrid, and Perovskite Photovoltaics XXV*; SPIE, 2024; p 1312307.
- (44) Horowitz, G. Organic Field-Effect Transistors. *Adv. Mater.* **1998**, *10*, 365–377.
- (45) Zaumseil, J.; Sirringhaus, H. Electron and Ambipolar Transport in Organic Field-Effect Transistors. *Chem. Rev.* **2007**, *107*, 1296–1323.
- (46) Sirringhaus, H. Device Physics of Solution-Processed Organic Field-Effect Transistors. *Adv. Mater.* **2005**, *17*, 2411–2425.
- (47) Babel, A.; Wind, J. D.; Jenekhe, S. A. Ambipolar Charge Transport in Air-Stable Polymer Blend Thin-Film Transistors. *Adv. Funct. Mater.* **2004**, *14*, 891–898.
- (48) Fratini, S.; Nikolka, M.; Salleo, A.; Schweicher, G.; Sirringhaus, H. Charge transport in high-mobility conjugated polymers and molecular semiconductors. *Nat. Mater.* **2020**, *19*, 491.
- (49) Lei, T.; Dou, J.-H.; Cao, X.-Y.; Wang, J.-Y.; Pei, J. Electron-Deficient Poly(p-phenylene vinylene) Provides Electron Mobility over 1 cm²/Vs under Ambient Conditions. *J. Am. Chem. Soc.* **2013**, *135*, 12168–12171.
- (50) Lei, T.; Xia, X.; Wang, J.-Y.; Liu, C.-J.; Pei, J. “Conformation Locked” Strong Electron-Deficient Poly(p-Phenylene Vinylene) Derivatives for Ambient-Stable n-Type Field-Effect Transistors: Synthesis, Properties, and Effects of Fluorine Substitution Position. *J. Am. Chem. Soc.* **2014**, *136*, 2135–2141.
- (51) Kang, B.; Kim, R.; Lee, S. B.; Kwon, S.-K.; Kim, Y.-H.; Cho, K. Side-Chain-Induced Rigid Backbone Organization of Polymer Semiconductors through Semifluoroalkyl Side Chains. *J. Am. Chem. Soc.* **2016**, *138*, 3679–3686.
- (52) Zhao, Z.; Yin, Z.; Chen, H.; Zheng, L.; Zhu, C.; Zhang, L.; Tan, S.; Wang, H.; Guo, Y.; Tang, Q.; Liu, Y. High-Performance, Air-Stable Field-Effect Transistors Based on Heteroatom-Substituted Naphtha-

lenediimide-Benzothiadiazole Copolymers Exhibiting Ultrahigh Electron Mobility up to 8.5 cm²/Vs. *Adv. Mater.* **2017**, *29*, 1602410.

(53) Chen, Z.; Zhang, W.; Huang, J.; Gao, D.; Wei, C.; Lin, Z.; Wang, L.; Yu, G. Fluorinated Dithienylethene-Naphthalenediimide Copolymers for High-Mobility n-Channel Field-Effect Transistors. *Macromolecules* **2017**, *50*, 6098–6107.

(54) Wang, Y.; Hasegawa, T.; Matsumoto, H.; Michinobu, T. Significant Improvement of Unipolar n-Type Transistor Performances by Manipulating the Coplanar Backbone Conformation of Electron-Deficient Polymers via Hydrogen Bonding. *J. Am. Chem. Soc.* **2019**, *141*, 3566–3575.

(55) Sheats, J. R.; Antoniadis, H.; Hueschen, M.; Leonard, W.; Miller, J.; Moon, R.; Roitman, D.; Stocking, A. Organic Electroluminescent Devices. *Science* **1996**, *273*, 884–888.

(56) Scholz, S.; Kondakov, D.; Lüssem, B.; Leo, K. Degradation Mechanisms and Reactions in Organic Light-Emitting Devices. *Chem. Rev.* **2015**, *115*, 8449–8503.

(57) Sirringhaus, H. Reliability of Organic Field-Effect Transistors. *Adv. Mater.* **2009**, *21*, 3859–3873.

(58) Iqbal, H. F.; Waldrip, M.; Chen, H.; McCulloch, I.; Jurchescu, O. D. Elucidating the Role of Water-Related Traps in the Operation of Polymer Field-Effect Transistors. *Adv. Electron. Mater.* **2021**, *7*, 2100393.

(59) Iqbal, H. F.; Ai, Q.; Thorley, K. J.; Chen, H.; McCulloch, I.; Risko, C.; Anthony, J. E.; Jurchescu, O. D. Suppressing bias stress degradation in high performance solution processed organic transistors operating in air. *Nat. Commun.* **2021**, *12*, 2352.

(60) Park, S.; Choi, W.; Kim, S. H.; Lee, H.; Cho, K. Protonated Organic Semiconductors: Origin of Water-Induced Charge-Trap Generation. *Adv. Mater.* **2023**, *35*, 2303707.

(61) Mateker, W. R.; McGehee, M. D. Progress in Understanding Degradation Mechanisms and Improving Stability in Organic Photovoltaics. *Adv. Mater.* **2017**, *29*, 1603940.

(62) Zhang, Q.; Chen, Y.; Liu, X.; Fahlman, M. In situ near-ambient pressure X-ray photoelectron spectroscopy reveals the effects of water, oxygen and light on the stability of PM6:Y6 photoactive layers. *J. Mater. Chem. C* **2023**, *11*, 3112–3118.

(63) Shoaee, S.; Luong, H. M.; Song, J.; Zou, Y.; Nguyen, T.-Q.; Neher, D. What We have Learnt from PM6:Y6. *Adv. Mater.* **2024**, *36*, 2302005.

(64) Tang, C. G.; Hou, K.; Leong, W. L. The Quest for Air Stability in Organic Semiconductors. *Chem. Mater.* **2024**, *36*, 28–53.

(65) Griggs, S.; Marks, A.; Bristow, H.; McCulloch, I. n-Type organic semiconducting polymers: stability limitations, design considerations and applications. *J. Mater. Chem. C* **2021**, *9*, 8099–8128.

(66) Kim, F. S.; Hwang, D.-K.; Kippelen, B.; Jenekhe, S. A. Enhanced carrier mobility and electrical stability of n-channel polymer thin film transistors by use of low-k dielectric buffer layer. *Appl. Phys. Lett.* **2011**, *99*, 173303.

(67) Briseno, A. L.; Kim, F. S.; Babel, A.; Xia, Y.; Jenekhe, S. A. n-Channel polymer thin film transistors with long-term air-stability and durability and their use in complementary inverters. *J. Mater. Chem.* **2011**, *21*, 16461–16466.

(68) Waldrip, M.; Jurchescu, O. D.; Gundlach, D. J.; Bittle, E. G. Contact Resistance in Organic Field-Effect Transistors: Conquering the Barrier. *Adv. Funct. Mater.* **2020**, *30*, 1904576.

(69) Lampert, Z. A.; Haneef, H. F.; Anand, S.; Waldrip, M.; Jurchescu, O. D. Tutorial: Organic field-effect transistors: Materials, structure and operation. *J. Appl. Phys.* **2018**, *124*, 071101.

(70) Tran, D. K.; West, S. M.; Guo, J.; Chen, S. E.; Ginger, D. S.; Jenekhe, S. A. Chain Length Dependence of Electron Transport in an n-Type Conjugated Polymer with a Rigid-Rod Chain Topology. *J. Am. Chem. Soc.* **2024**, *146*, 1435–1446.

(71) Phan, H.; Ford, M. J.; Lill, A. T.; Wang, M.; Bazan, G. C.; Nguyen, T.-Q. Electrical Double-Slope Nonideality in Organic Field-Effect Transistors. *Adv. Funct. Mater.* **2018**, *28*, 1707221.

(72) Kalb, W. L.; Batlogg, B. Calculating the trap density of states in organic field-effect transistors from experiment: A comparison of different methods. *Phys. Rev. B* **2010**, *81*, 035327.

(73) Haneef, H. F.; Zeidell, A. M.; Jurchescu, O. D. Charge carrier traps in organic semiconductors: a review on the underlying physics and impact on electronic devices. *J. Mater. Chem. C* **2020**, *8*, 759–787.

(74) Feng, K.; Zhang, X.; Wu, Z.; Shi, Y.; Su, M.; Yang, K.; Wang, Y.; Sun, H.; Min, J.; Zhang, Y.; Cheng, X.; Woo, H. Y.; Guo, X. Fluorine-Substituted Dithienylbenzodiiimide-Based n-Type Polymer Semiconductors for Organic Thin-Film Transistors. *ACS Appl. Mater. Interfaces* **2019**, *11*, 35924–35934.

(75) Zhang, T.; Chen, Z.; Zhang, W.; Wang, L.; Yu, G. Recent Progress of Fluorinated Conjugated Polymers. *Adv. Mater.* **2024**, *36*, 2403961.

(76) Furukawa, S.; Yasuda, T. Strategic end-halogenation of π -conjugated small molecules enabling fine morphological control and enhanced performance of organic solar cells. *J. Mater. Chem. A* **2019**, *7*, 14806–14815.

(77) Yang, F.; Li, C.; Lai, W.; Zhang, A.; Huang, H.; Li, W. Halogenated conjugated molecules for ambipolar field-effect transistors and non-fullerene organic solar cells. *Mater. Chem. Front.* **2017**, *1*, 1389–1395.

(78) Peng, J.; Ma, L.; Li, H.; Wang, G.; Chen, Z.; Chen, F.; Hou, J.; Zhang, S. A comprehensive study on the halogenation effect of non-fullerene acceptors for photovoltaic application. *Mater. Chem. Front.* **2024**, *8*, 3643.

(79) Zhang, H.; Yao, H.; Hou, J.; Zhu, J.; Zhang, J.; Li, W.; Yu, R.; Gao, B.; Zhang, S.; Hou, J. Over 14% Efficiency in Organic Solar Cells Enabled by Chlorinated Nonfullerene Small-Molecule Acceptors. *Adv. Mater.* **2018**, *30*, 1800613.

(80) Feng, K.; Guo, H.; Wang, J.; Shi, Y.; Wu, Z.; Su, M.; Zhang, X.; Son, J. H.; Woo, H. Y.; Guo, X. Cyano-Functionalized Bithiophene Imide-Based n-Type Polymer Semiconductors: Synthesis, Structure-Property Correlations, and Thermoelectric Performance. *J. Am. Chem. Soc.* **2021**, *143*, 1539–1552.

(81) Casey, A.; Dimitrov, S. D.; Shakya-Tuladhar, P.; Fei, Z.; Nguyen, M.; Han, Y.; Anthopoulos, T. D.; Durrant, J. R.; Heeney, M. Effect of Systematically Tuning Conjugated Donor Polymer Lowest Unoccupied Molecular Orbital Levels via Cyano Substitution on Organic Photovoltaic Device Performance. *Chem. Mater.* **2016**, *28*, 5110–5120.

(82) Chen, Z.; Li, J.; Wang, J.; Yang, K.; Zhang, J.; Wang, Y.; Feng, K.; Li, B.; Wei, Z.; Guo, X. Imide-Functionalized Fluorenone and Its Cyanated Derivative Based n-Type Polymers: Synthesis, Structure-Property Correlations, and Thin-Film Transistor Performance. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202205315.

(83) Feng, K.; Shan, W.; Wang, J.; Lee, J.-W.; Yang, W.; Wu, W.; Wang, Y.; Kim, B. J.; Guo, X.; Guo, H. Cyano-Functionalized n-Type Polymer with High Electron Mobility for High-Performance Organic Electrochemical Transistors. *Adv. Mater.* **2022**, *34*, 2201340.

(84) Shi, Y.; Guo, H.; Qin, M.; Wang, Y.; Zhao, J.; Sun, H.; Wang, H.; Wang, Y.; Zhou, X.; Facchetti, A.; Lu, X.; Zhou, M.; Guo, X. Imide-Functionalized Thiazole-Based Polymer Semiconductors: Synthesis, Structure-Property Correlations, Charge Carrier Polarity, and Thin-Film Transistor Performance. *Chem. Mater.* **2018**, *30*, 7988–8001.

(85) Onwubiko, A.; Yue, W.; Jellett, C.; Xiao, M.; Chen, H.-Y.; Ravva, M. K.; Hanifi, D. A.; Knall, A.-C.; Purushothaman, B.; Nikolka, M.; Flores, J.-C.; Salleo, A.; Bredas, J.-L.; Sirringhaus, H.; Hayoz, P.; McCulloch, I. Fused electron deficient semiconducting polymers for air stable electron transport. *Nat. Commun.* **2018**, *9*, 416.

(86) West, S. M.; Tran, D. K.; Guo, J.; Chen, S. E.; Ginger, D. S.; Jenekhe, S. A. Phenazine-Substituted Poly-(benzimidazobenzophenanthroline-dione): Electronic Structure, Thin Film Morphology, Electron Transport, and Mechanical Properties of an n-Type Semiconducting Ladder Polymer. *Macromolecules* **2023**, *56*, 2081–2091.

(87) Tran, D. K.; West, S. M.; Speck, E. M. K.; Jenekhe, S. A. Observation of super-Nernstian proton-coupled electron transfer and elucidation of nature of charge carriers in a multiredox conjugated polymer. *Chem. Sci.* **2024**, *15*, 7623–7642.

(88) Izuhara, D.; Swager, T. M. Poly(Pyridinium Phenylene)s: Water-Soluble N-Type Polymers. *J. Am. Chem. Soc.* **2009**, *131*, 17724–17725.